

Science Reporting Summary — item-by-item answers

Fill the journal's online form from this sheet. Values are from `paper/tables.md` (generated) and

`results/CORRECTION.md` (the corrected experiment). Anything marked **[you]** is author-side.

Study design

item	answer
Study type	Computational experiment on a connectome-constrained spiking model. No animals, no human participants, no clinical data.
Unit of analysis	One simulated trial (independent stimulus stream and injection draws).
n per group	10 trials per arm (I2-heading, M2-dose, A4-flow, D2-controls); 20 per arm in the replication (J2-n20); 6 per level at the two extra gains (K2-k150, L2-k600).
Randomisation	Arm order is fixed by design; every arm draws its injections from its own <code>Random</code> stream, and all arms in a trial share one stimulus stream (<code>Random(90000 + trial)</code>), so the stimulus is bit-identical across arms.
Blinding	Not applicable — no subjective measurement. Every readout is computed by code from spike counts.
Inclusion / exclusion	No trials excluded. <code>n_excluded_nonfinite</code> is reported per metric in <code>results/raw/*-compare.csv</code> (0 for every reported metric).
Replication	Internal: the whole experiment was re-run after the circuit was rebuilt (see <code>results/CORRECTION.md</code>), and the conflict dose was replicated at $\kappa = 150$ and $\kappa = 600$.
Sample-size justification	The Bayes-factor ceiling of the design is set by n , not by the analysis: $\text{BF}_{01}(t = 0) = 2.52$ at $n = 10$ and 3.24 at $n = 20$ (<code>tools/make_main_figures.py: supp3</code>). $n = 20$ was chosen to cross $\text{BF} = 3$ for the primary readouts.

Data and materials

item	answer
Source data	MaleCNS v1.0 flat connectome (Google × Janelia / FlyWire Consortium; male CNS, brain + VNC). Exact files, byte counts and SHA256 in <code>paper/supplementary-note-S1.md §S1.1</code> .
Derived data	<code>flymind.brain (FLYMIND4)</code> , SHA256 <code>FC6B544C68B4C9997987E2BAA788A41ED8E4E91B3B5C60EBC1166947447827AB</code> ; reconstruction command in <code>reproduce.ps1 -Tier source</code> .
New reagents	None. This is a computational study.
Materials availability	Not applicable.

Statistics

item	answer
Tests	Welch t with 95% CI; Cohen's d with Hedges–Olkin CI; Mann–Whitney U with tie correction and rank-biserial correlation; permutation tests (4,000 relabellings, seed 12345); Holm–Bonferroni within each metric's self-vs-

	control family.	
Equivalence	TOST at two pre-specified bounds: standardised (λ	$d \leq 0.5$) and relative ($\pm 10\%$ of the control mean), with 90% CIs.
Bayes factors	JZS two-sample, Cauchy prior scale $r = 0.707$; validated against the analytic anchor $BF_{01}(t = 0, n = 10/\text{arm}) = 2.52$ (python <code>tools/compare_trials.py --selftest</code>).	
Multiplicity	Reported per contrast as raw P plus Holm-corrected P, and read against Bayes factors; no metric is claimed on corrected P alone.	
Error bars in figures	Mean $\pm$ 95% CI everywhere; individual trial values are overlaid as points.	
Analysis software	Python 3.14.7 (standard library + numpy/matplotlib for figures), .NET 8.0.404 for the simulator. No statistical package is used: every test is implemented in <code>tools/compare_trials.py</code> .	

Reproducibility

item	answer		
One-command reproduction	<code>`pwsh -File reproduce.ps1 -Tier sims\`</code>	stats\	figures`
Blob check	<code>python tools\verify_headline_numbers.py --25</code> checks, all passing against the deposited CSVs		
Environment	Windows, .NET 8.0.404, Python 3.14.7; software versions in <code>results/PROVENANCE.md</code>		
Compute	One workstation, single-threaded simulator, $0.81\times$ real time; the full family set is ~ 25 min wall-clock with 8 processes in parallel		

Data, code and AI

item	answer
Data availability	Repository URL: [you — fill after pushing] ; the blob, every raw per-trial CSV and every stdout log are included. The 3 GB edge list and the 1 GB source feathers are <i>not</i> redistributed; they are rebuilt from the released MaleCNS v1.0 files by <code>tools/export_edges.py</code> (see <code>paper/supplementary-note-S1.md</code>).
Code availability	Same repository; MIT for code, CC BY 4.0 for data and figures (see <code>LICENSE</code>).
Generative AI — text	Disclosed: Methods (<i>Use of generative AI</i>), Acknowledgments, cover letter, and the full prompt log in Supplementary Note S2.
Generative AI — images	None. Every figure is a programmatic plot of the authors' own numerical output (matplotlib scripts in <code>tools/</code>); no AI image tool was used.
Generative AI — code	The model, analysis and figure code was written with a large language model under author direction; all of it was executed and its output verified by the authors (Note S2).
Dual use / biosafety	None.
Third-party material	None; no previously published figure or image is reproduced.